

Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and preferences across **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

27%

Subscribers

Features span demographics, purchase details, and shopping behavior.

Data Preparation in Python

→ Clean & Standardize

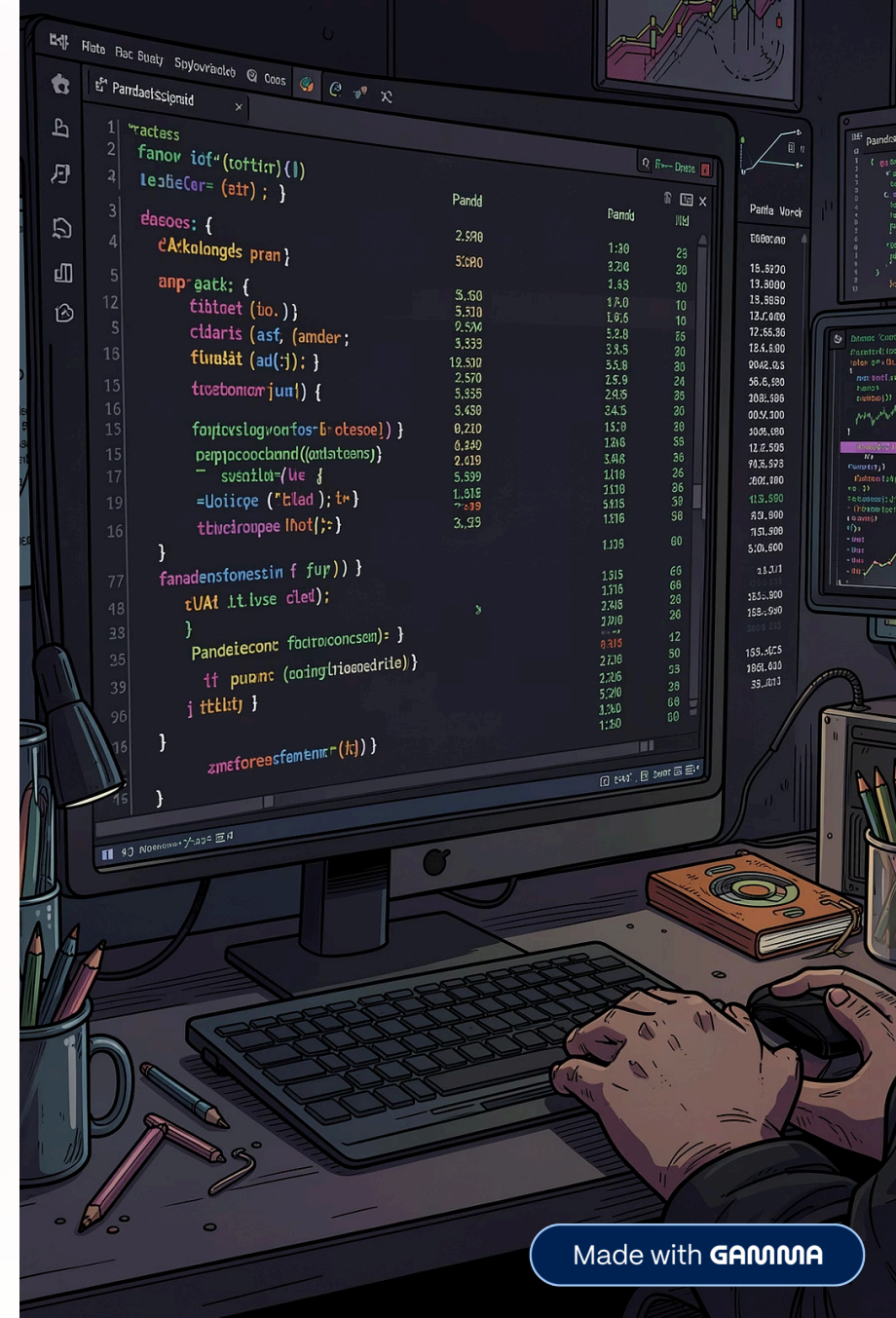
Imputed missing Review Ratings; renamed columns to snake_case

→ Feature Engineering

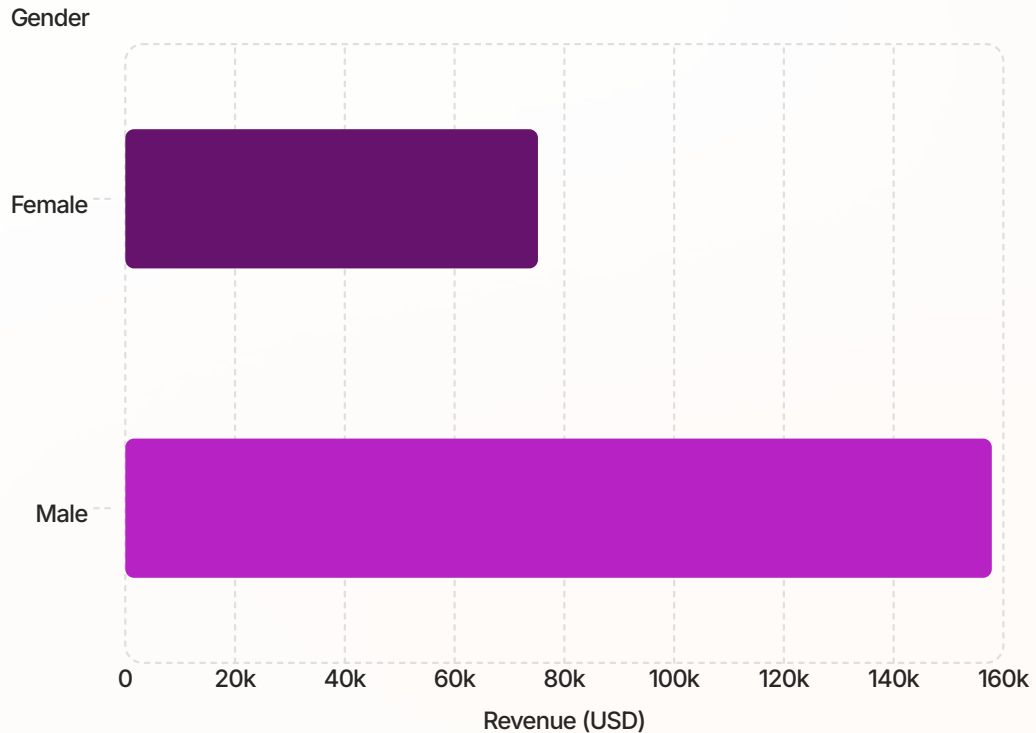
Created **age_group** and **purchase_frequency_days**

→ Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis



Revenue by Gender



Male customers drive 2× more revenue

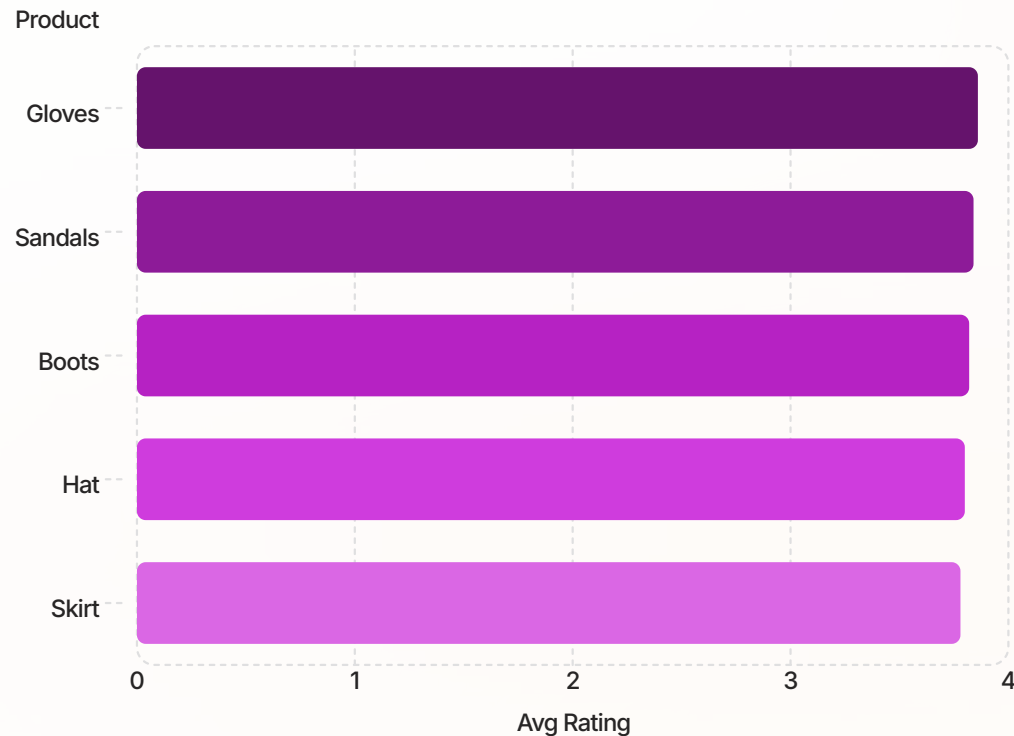
Male: **\$157,890**

Female: **\$75,191**

Male shoppers also represent 68% of all customers (2,652 vs. 1,248).

Top Products & Discount Patterns

Top 5 Products by Rating

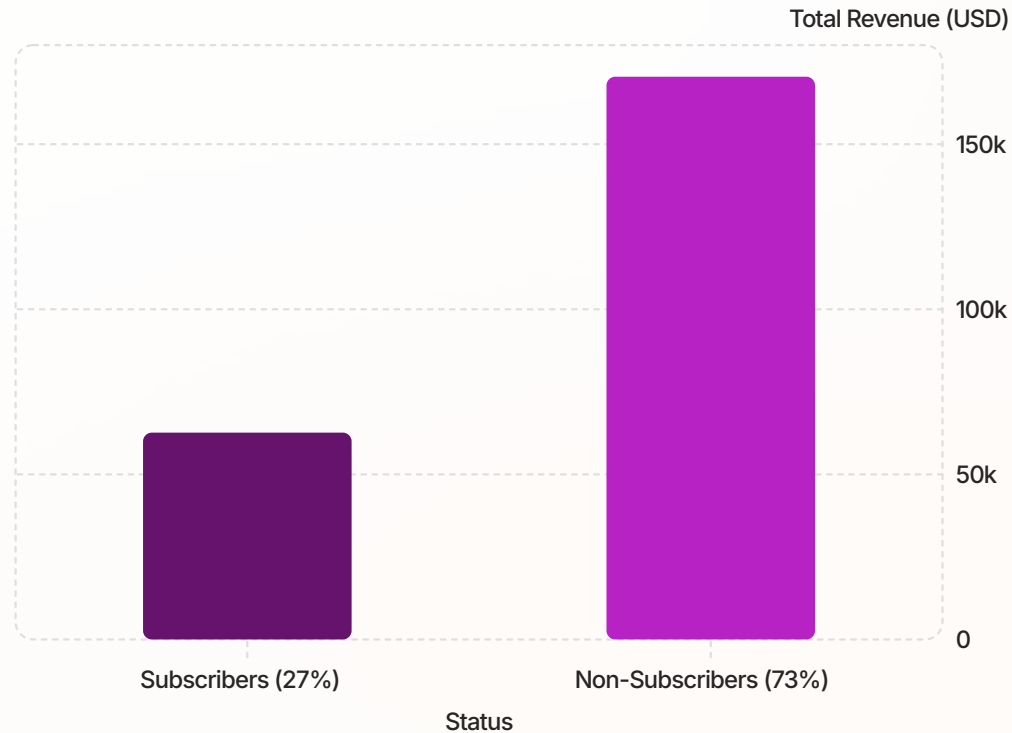


Most Discounted Products

- Hat — 50.00%
- Sneakers — 49.66%
- Coat — 49.07%
- Sweater — 48.17%
- Pants — 47.37%

839 high-spending customers used discounts above average purchase amount.

Subscribers vs. Non-Subscribers



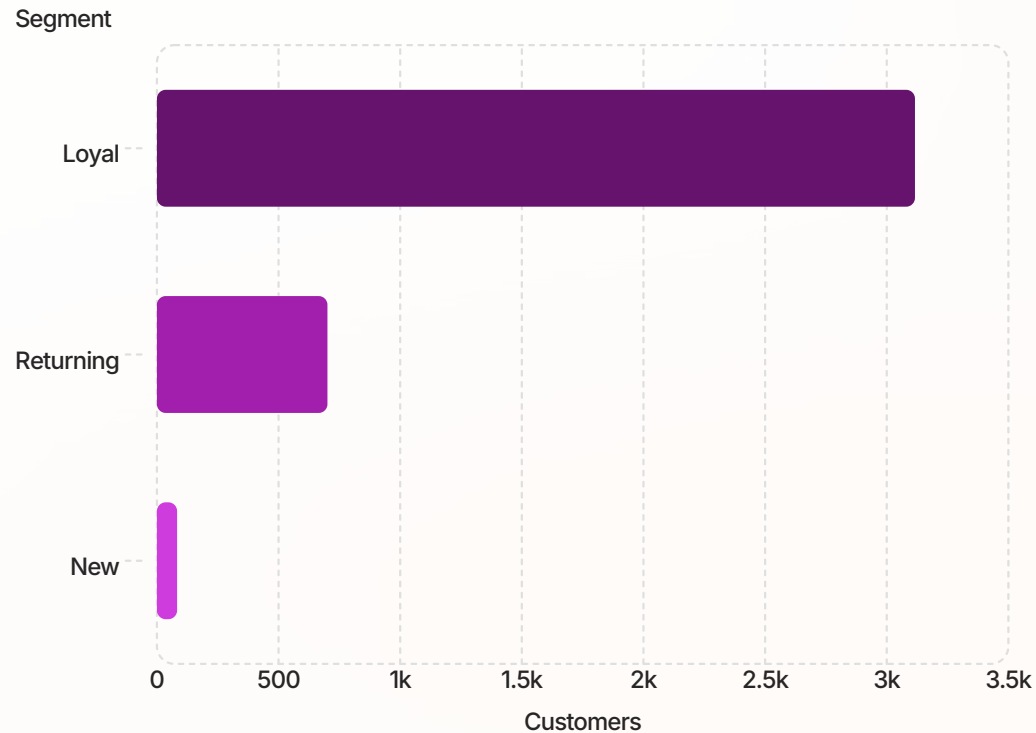
Non-subscribers generate 2.7× more revenue

Average spend is nearly identical:

Subscribers: **\$59.49** · Non-Subscribers: **\$59.87**

However, 958 repeat buyers (>5 purchases) are subscribers vs. 2,518 non-subscribers — suggesting room to convert.

Customer Segmentation



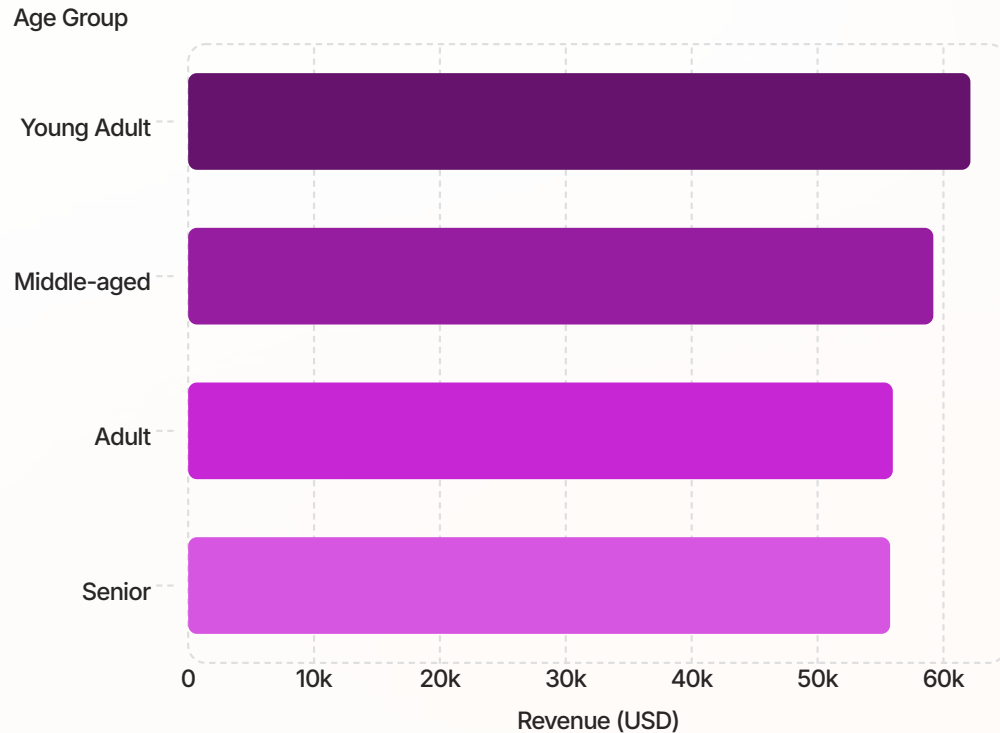
Loyal customers dominate

80% of customers are Loyal, yet only 958 of them subscribe.



Opportunity: Convert Loyal non-subscribers with exclusive benefits.

Revenue by Age Group



Young Adults lead revenue

Revenue is relatively evenly distributed across age groups, with Young Adults contributing the most at **\$62,143**.

Average purchase amount across all customers: **\$59.76**



Power BI Dashboard

Key Metrics

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription, Gender, Category,
Shipping Type

Visualizations

Revenue & Sales by Category and Age Group



Loyalty Programs

Review Discount Policy

Targeted Marketing

Focus on high-revenue age groups & express-shipping users